

**EARLY DETECTION OF LITHIUM
DENDRITE RISK IN EV BATTERIES
USING QUANTUM SUPPORT VECTOR
MACHINES**

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ABSTRACT

The overall number of electric vehicles (EVs) on the roads is increasing and the need for the safety and reliability of their batteries is becoming more important. Dendrite formation is one of the most critical issues facing an electric vehicle's battery and can cause not only safety hazards but also destruction of the battery itself. This project offers a unique procedure for spotting battery dendrite risk early, using a hybrid machine learning model that merges both classical and quantum computing methods. Quantum Support Vector Machines classify the health of the battery time-series data, which consists of voltage, current and temperature. The sensors that are placed on the battery gather continuous data for a certain period of time and this data is then divided into smaller segments using a sliding window technique of pre-processing. For each segment, various statistics including average, variance and rate of change are computed to identify potential features in the battery's behavior. Features that have been selected are subsequently provided to the hybrid model where the QSVM further pushes the feature transformation process making it easier to detect very slight abnormalities which are signs of dendrite growth. It is the objective of this model to generate real-time, very precise predictions of dendrite risk that will facilitate quick actions taken and thus better management of the overall battery. The fusion of quantum computing and traditional machine learning techniques is what marks a key milestone for the EVs battery health monitoring sector.

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LIST OF ABBREVIATIONS

QSVM Quantum Support Vector Machine

QML Quantum Machine Learning

SVM Support Vector Machine

EV Electric Vehicle

QAOA Quantum Approximate Optimization Algorithm

NISQ Noisy Intermediate-Scale Quantum

CHAPTER 1

INTRODUCTION

EVs can contribute to sustainable transportation and lithium-ion batteries are the key to their operation. The charging process subjects the EV batteries to different electrical and thermal conditions, which when allowed to get out of check may lead to abnormal behavior. Thermal runaway is one of the most important safety issues since the phenomenon, which is defined as an abrupt rise in temperature, may cause battery deterioration or system malfunction. Such abnormal conditions must therefore be identified early in order to be able to ensure that operations are safe and also to guarantee reliability of the system.

Due to the growing implementation of sensor-based battery management systems, nowadays it is possible to collect a considerable amount of time-stamped battery charging data in real time. This data is usually continuous measurements, including voltage, current, temperature and state of charge, measured during the charge, which are recorded during the charging cycle. This data is usually analyzed with machine learning methods and used to identify abnormal charging conduct. Nevertheless, most of the current methods are based on classical machine learning models only, and these might not be able to depict the complicated nonlinear relationships that can exist in multivariate time-series sensor data.

The new developments in quantum computing provided quantum-inspired machine learning (QML) methods that present alternative data representation and transformation techniques. Despite the fact that mod-

ern quantum devices have not offered any computational advantage of practical datasets, hybrid learning systems with classical and quantum-inspired components have been considered, because the feature representation can be improved. Here, it is assumed that QML will not replace the classical machine learning, but it represents another processing layer that can be added to the already existing learning pipeline.

This paper suggests a hybrid classical-quantum learning model to detect anomalies in EV battery charging systems with real time stamped sensor data. It is based on the suggested procedure that combines the classical time-series preprocessing and feature extraction with a quantum-inspired processing layer that is aimed at executing nonlinear feature transformation. The resulting hybrid feature representation is then studied with a classical classifier to determine abnormal charging patterns that relate to thermal runaway.

This project will have the main goal of designing and deploying an entire learning pipeline showing the extent to which advanced machine learning techniques can be integrated in a safety-critical and real world application. Instead of prioritizing the superior performance or computational acceleration, it focuses on the architectural design, feasibility, and the systematization of evaluation of hybrid feature representations of time-series anomaly detection. The efficiency of the suggested framework is assessed with the help of the data set of time-stamped sensor readings that were recorded in EV battery charge and thermal runaway events.

CHAPTER 2

LITERATURE REVIEW

Table 2.1 Literature Review

Paper	Methodology	Limitations	Ideas for Adoption
Comparative Investigation of Quantum and Classical Kernel Functions Applied in Support Vector Machine Algorithms	Performs a comparative study of classical kernels and quantum kernels within the SVM framework. Different quantum feature maps are evaluated using QSVM on benchmark datasets.	Uses quantum simulators; high kernel computation cost; limited scalability; lacks real-world application focus.	Adopt the quantum kernel comparison framework to evaluate classical SVM versus QSVM for battery time-series features, justifying improved separability for early dendrite-risk patterns.

Paper	Methodology	Limitations	Ideas for Adoption
Assessment of Quantum Feature Map for Binary Forest Classification using QSVM	Evaluates multiple Pauli-based quantum feature maps for binary classification. Features are normalized, reduced, encoded, and classified using QSVM with Pareto-front analysis.	Relies on simulators; inconsistent QSVM superiority; scalability limitations in NISQ-era hardware.	Apply quantum feature map selection strategies to battery sensor data and use Pareto analysis to balance accuracy and computational cost.
Hybrid Quantum–Classical Framework for Urban Traffic State Classification	Implements a hybrid pipeline combining classical feature extraction with quantum encoding and QSVM-based classification.	Limited dataset; simulator-based execution; real-time scalability not addressed.	Use the hybrid classical–quantum architecture for battery monitoring by applying QSVM after classical preprocessing of voltage and temperature data.

Paper	Methodology	Limitations	Ideas for Adoption
AutoQML: A Framework for Automated Quantum Machine Learning	Introduces automated quantum model selection, feature encoding, and hyperparameter tuning in hybrid workflows.	High computational overhead; simulator dependence; not tailored for time-series data.	Adopt automated quantum kernel and hyperparameter tuning to reduce manual QSVM optimization for battery datasets.
Real-Time Quantum Machine Learning-Based Anomaly Detection for Lithium-Ion Battery Packs	Proposes quantum-enhanced K-Means clustering for real-time anomaly detection using battery sensor data.	Focuses on clustering; requires multiple qubits; real-time hardware feasibility unclear.	Extend the anomaly detection framework by replacing clustering with QSVM-based supervised dendrite-risk classification.
Quantum Kernel-Based Support Vector Machine with QAOA Embedding for Lung Cancer Prediction	Uses QAOA-based quantum feature embedding with QSVM, achieving better results for small datasets.	Simulator execution; high cost; evaluated only on static medical data.	Adopt QAOA-based quantum embedding for capturing nonlinear patterns in reduced battery feature sets.

CHAPTER 3

SYSTEM DESIGN

3.1 DESIGN OVERVIEW

The proposed system design aims at developing a dependable hybrid Quantum Support Vector Machine (QSVM)-based system for detecting anomalies and assessing risks in electric vehicle (EV) battery charging systems. The architecture is inspired by existing quantum machine learning-based battery anomaly detection solutions, particularly those based on unsupervised clustering algorithms such as K-means.

Although clustering-based methods can identify coarse deviations in battery behaviour, they often fail to detect subtle early-stage anomalies and lack clear decision margins. To overcome these limitations, the proposed system replaces clustering with a margin-based hybrid QSVM classifier combined with a risk-oriented decision layer.

The system follows a modular pipeline architecture in which each module performs a specific transformation and passes its output to the next stage. This design improves interpretability, scalability, and ease of integration with existing battery management systems.

3.2 SYSTEM OVERVIEW

The EV battery time-stamped sensor data is processed through the following modules:

- Data Acquisition
- Data Preprocessing

- Classical Feature Engineering
- Quantum-Inspired Feature Mapping
- Hybrid QSVM Classification
- Risk Assessment and Output Monitoring

Data preparation and baseline feature extraction are performed using classical machine learning techniques. Quantum-inspired transformations are then used to enhance the feature representation before classification using a QSVM model.

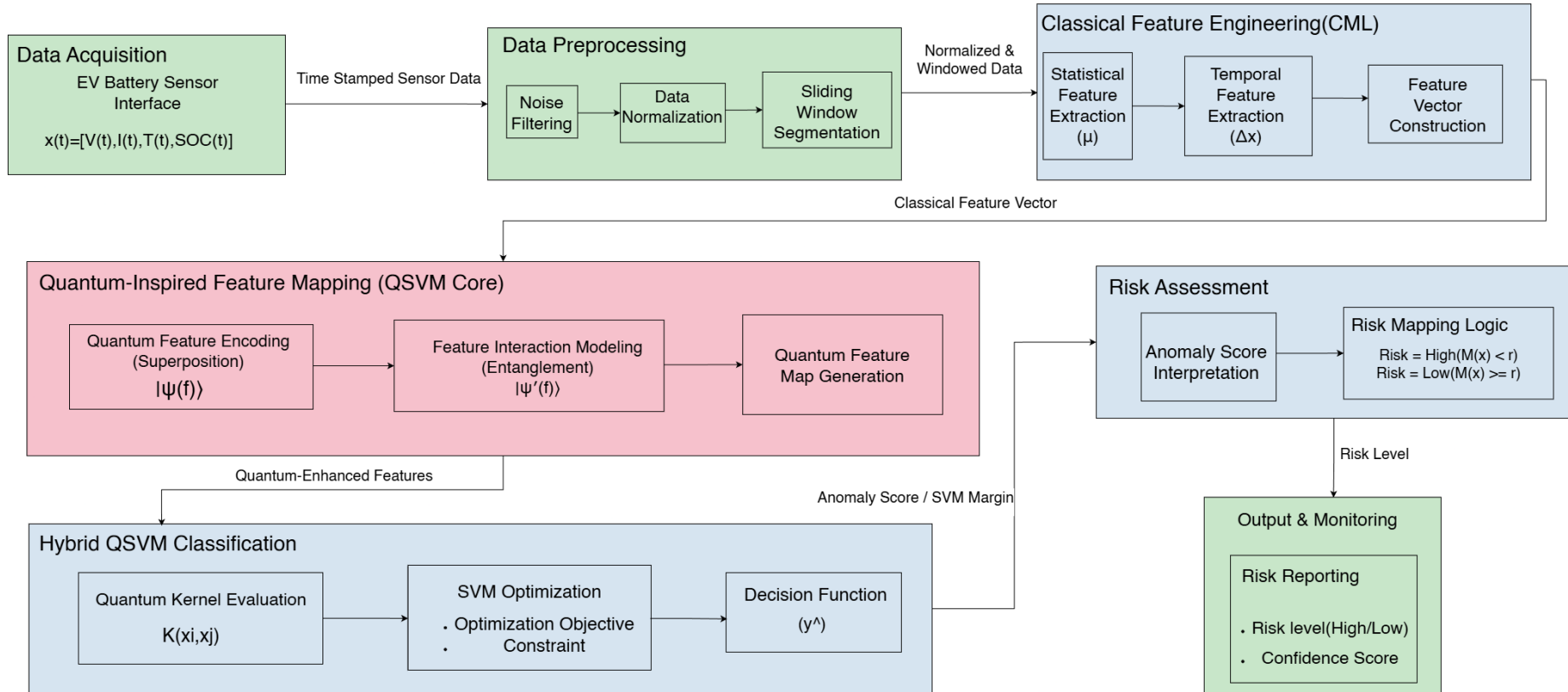


Figure 3.1 Architecture of the Proposed QSVM-Based EV Battery System

3.3 SYSTEM OVERVIEW

3.4 DATA ACQUISITION MODULE

The data acquisition module gathers time-stamped sensor data from EV battery packs during charging cycles. Each record contains synchronized measurements of:

- Voltage
- Current
- Temperature
- State of Charge (SOC)

These parameters collectively form a multivariate time-series representation of battery behaviour. Since battery anomalies often result from interactions between multiple parameters rather than single threshold violations, collecting multiple sensor signals provides a more comprehensive monitoring framework.

Algorithm 1 Battery Data Loading Module

Input: NASA battery .mat files

Output: Structured dataset D

for each battery file B_i **do**

Load B_i using MATLAB reader

Extract cycle structure

Separate discharge, charge, impedance cycles

Store structured data in D

end for

return D

3.5 DATA PREPROCESSING MODULE

Raw sensor data may contain noise, outliers, and variations in measurement scales. Therefore, preprocessing is necessary before feature extraction.

The preprocessing module performs:

- Noise filtering
- Data normalization
- Sliding window segmentation

Noise filtering reduces sensor disturbances, normalization ensures comparable feature ranges, and sliding window segmentation divides the time-series data into fixed-length windows that represent short-term charging behaviour.

Algorithm 2 Sliding Window Degradation Modeling

Input: Feature matrix F
Output: Windowed dataset W
Define window size w and stride s
for each battery b **do**
 for each window segment **do**
 Compute capacity trend (slope)
 Compute capacity change (last - first)
 Compute impedance trend
 Compute degradation variance
 Store window features
 end for
end for
return W

3.6 CLASSICAL FEATURE ENGINEERING MODULE

After preprocessing, statistical feature engineering is applied to extract meaningful descriptors from each window.

Typical features include:

- Mean
- Variance
- Rate of change
- Stability indicators

These features form a structured classical feature vector represent-

ing the battery behaviour within a time window. This stage reduces the dimensionality of raw sensor data and provides interpretable inputs for further analysis.

Algorithm 3 Statistical and Electrochemical Feature Extraction

Input: Dataset D

Output: Feature matrix F

for each discharge cycle c in D **do**

 Extract capacity

 Compute voltage statistics (mean, std, min, max, range)

 Compute current statistics (mean, std)

 Compute temperature statistics (mean, std, min, max)

 Extract impedance parameters (Re, Rct)

end for

Compute Coulombic Efficiency:

$$CE = \frac{Q_{discharge}}{Q_{charge}}$$

Merge all features into F

return F

3.7 QUANTUM-INSPIRED FEATURE MAPPING MODULE

The quantum-inspired feature mapping module converts classical feature vectors into a higher-dimensional Hilbert space using quantum encoding techniques.

Initially, classical features are encoded using superposition-based representations, allowing multiple components to be represented simultaneously. Entanglement-inspired transformations are then applied to capture interactions between features.

These transformations increase the expressive capability of the feature space without requiring physical quantum hardware.

3.8 HYBRID QSVM CLASSIFICATION MODULE

The proposed system uses a hybrid QSVM classifier to detect anomalous charging behaviour.

Unlike clustering-based approaches, QSVM learns a margin-based decision boundary that separates normal and abnormal battery patterns in the transformed feature space.

The similarity between feature vectors is computed using a quantum kernel:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

The QSVM combines quantum kernel feature mapping with classical SVM optimization.

3.9 RISK ASSESSMENT AND OUTPUT GENERATION

The system includes a risk assessment module that interprets the QSVM output to classify charging behaviour into risk levels.

The anomaly score or decision margin produced by the QSVM is compared with predefined thresholds to categorize each sample as:

- High Risk
- Low Risk

The final system outputs include:

- Risk level
- Confidence score
- Monitoring reports

3.10 DESIGN SUMMARY

The proposed system extends existing battery anomaly detection frameworks by incorporating quantum-inspired feature mapping within a classical machine learning pipeline. The use of a hybrid QSVM

classifier enables better separation between normal and abnormal battery behaviour while maintaining interpretability and computational efficiency.

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 OVERVIEW OF THE PROPOSED ARCHITECTURE

The proposed framework is organized as a sequence of interconnected modules, each responsible for a specific stage in battery degradation analysis.

The system combines classical signal processing techniques with quantum-enhanced machine learning to predict battery risk levels.

The pipeline consists of the following modules:

1. Environment Setup
2. Data Loading
3. Timestamp Alignment
4. Missing Value Handling
5. Statistical Feature Extraction
6. Dendrite-Oriented Feature Extraction
7. Sliding Window Modelling
8. Window-Level Feature Construction
9. Quantum Kernel SVM Classification
10. Risk Assessment and Reporting

4.2 ENVIRONMENT SETUP

This module prepares the computational environment required for implementation. All required libraries for numerical computation, visualization, machine learning, and MATLAB file processing are imported.

The dataset stored in Google Drive is mounted to ensure direct access during execution.

4.3 DATA LOADING

The NASA battery dataset stored in MATLAB format is loaded in this stage. A total of 35 battery files were available, out of which 33 valid files were successfully processed.

Each file contains multiple cycles including:

- Charge cycles
- Discharge cycles
- Impedance measurements

In total, 7,313 cycles were loaded across all batteries.

4.4 TIMESTAMP ALIGNMENT

Battery cycles contain associated time arrays. These timestamps are aligned with cycle indices to ensure consistent temporal comparisons across samples.

4.5 MISSING VALUE HANDLING

Different cycle types contain different measurements. The following parameters are extracted safely:

- Capacity
- Voltage
- Current
- Temperature
- Time
- Impedance components (Re and Rct)

A total of 4,600 discharge and impedance cycles were extracted. Remaining missing values were handled using median-based imputation.

4.6 STATISTICAL FEATURE ENGINEERING

Time-series signals are converted into numerical descriptors for each discharge cycle.

The following statistical features are computed:

- Mean
- Standard deviation
- Minimum
- Maximum
- Range

Additional impedance characteristics such as R_e and R_{ct} are also included.

After filtering invalid entries, 2,667 valid cycles remained, producing 17 statistical features.

4.7 DENDRITE-SPECIFIC FEATURE ENGINEERING

Additional electrochemical features related to dendrite formation were extracted.

Differential capacity is computed as:

$$\frac{dQ}{dV}$$

Other features include:

- Skewness
- Kurtosis
- Entropy
- Interquartile range

Coulombic efficiency is calculated as:

$$CE = \frac{Q_{discharge}}{Q_{charge}}$$

After merging all descriptors, the dataset contained 18 engineered features.

4.8 SLIDING WINDOW MODELLING

Battery degradation exhibits temporal behaviour. Therefore sliding windows were applied across cycle sequences.

Window sizes used:

- 5 cycles
- 10 cycles
- 20 cycles

Stride size: 6 cycles

This process generated 352 window-level samples across the 33 batteries.

4.9 WINDOW-LEVEL FEATURE CONSTRUCTION

Within each sliding window, trend features were computed including:

- Capacity mean
- Capacity standard deviation
- Capacity slope
- Capacity change
- Degradation variance

A total of 33 window-level features were produced.

4.10 QUANTUM FEATURE MAPPING WITH PCA

Principal Component Analysis (PCA) was applied to reduce dimensionality before quantum encoding.

The feature space was reduced from 31 features to 4 principal components, scaled to the range $[0, \pi]$ for quantum circuit rotations.

4.11 QUANTUM KERNEL SVM CLASSIFICATION

A 4-qubit quantum circuit was constructed.

Operations used:

- Hadamard gates
- RZ rotations
- CNOT entanglement

The quantum state is represented as:

$$|\psi(x)\rangle = U_\phi(x)|0\rangle$$

The kernel similarity is computed as:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

The dataset was split into:

- Training: 281 samples
- Testing: 71 samples

The final QSVM model contained 52 support vectors.

4.12 TEST CASES

4.12.1 Dataset and Feature Summary

4.12.2 Model Performance

4.13 RESULTS AND DISCUSSION

The experimental results demonstrate that the hybrid QSVM model successfully identifies battery degradation patterns and predicts risk levels.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 SUMMARY OF THE WORK

This powerful data processing pipeline established a cornerstone. We successfully converted raw, heterogeneous battery data into a standardized, clean, and feature-rich dataset. These multi-scale temporal features now prepare for the next phase of this project: the development and training of Quantum Machine Learning models for predictive analytics, with the main focus on early detection and understanding of battery dendrite formation to enhance battery safety, life, and performance. Feature vectors generated at this step are the required input for advanced machine learning algorithms and allow the study of subtle signatures of degradation that might indicate dendrite growth.

5.2 QUANTUM MACHINE LEARNING ENHANCEMENTS

Future improvements to the system can enhance quantum feature representation and classification performance.

5.2.1 Expand Qubit Count

Increasing the number of qubits from 4 to 6–8 would expand the feature space to 64–256 dimensions.

Maintaining PCA variance between 80–90% may improve kernel expressiveness and reduce support vector ratios.

5.2.2 Advanced Quantum Feature Maps

Future implementations may replace the existing feature map with advanced circuits such as:

- Pauli Feature Maps
- IQP circuits

Trainable quantum kernels can also be introduced to maximize class separability.

5.2.3 Hybrid Variational Quantum Circuits

Variational Quantum Eigensolver (VQE) or Quantum Neural Networks may be integrated before the SVM classifier to learn latent quantum representations.

5.3 CLASSICAL MACHINE LEARNING ENHANCEMENTS

5.3.1 Improved Label Design

Future work may incorporate physics-based thresholds such as State of Health (SOH) limits instead of statistical quantile labels.

5.3.2 Enhanced Feature Engineering

Additional electrochemical features such as Electrochemical Impedance Spectroscopy (EIS) parameters may be included.

Wavelet transforms can also be used to extract multi-frequency degradation signatures.

5.3.3 Classical Baseline Comparison

Benchmarking against classical models such as:

- Random Forest
- XGBoost
- LSTM

can help evaluate the practical advantages of quantum models.

5.4 DATA AND SCALABILITY

Future experiments may include larger datasets from NASA and CALCE battery repositories to improve generalization.

Testing on real quantum hardware using IBM Quantum devices can also evaluate practical quantum advantage.

5.5 SYSTEM INTEGRATION AND DEPLOYMENT

The risk prediction module can be integrated into Battery Management Systems (BMS) as a REST API microservice.

Real-time dashboards can visualize battery risk levels and trigger alerts when critical thresholds are reached.

Explainability methods such as SHAP can be used to interpret feature contributions in the model predictions.

5.6 MULTI-CHEMISTRY EXTENSION

Future systems may extend the methodology to other battery chemistries such as lithium-sulfur or solid-state batteries, enabling broader battery safety monitoring applications.